solenoid valve through relay RL1. A 1N4007 diode (D1) is used as flywheel diode. The solenoid valve is connected to the common terminal of relay RL1. It gets 12V DC power source through NC contact of the relay.

The solenoid valve is placed at cold-water inlet of the solar water heater system to control the flow of incoming water.

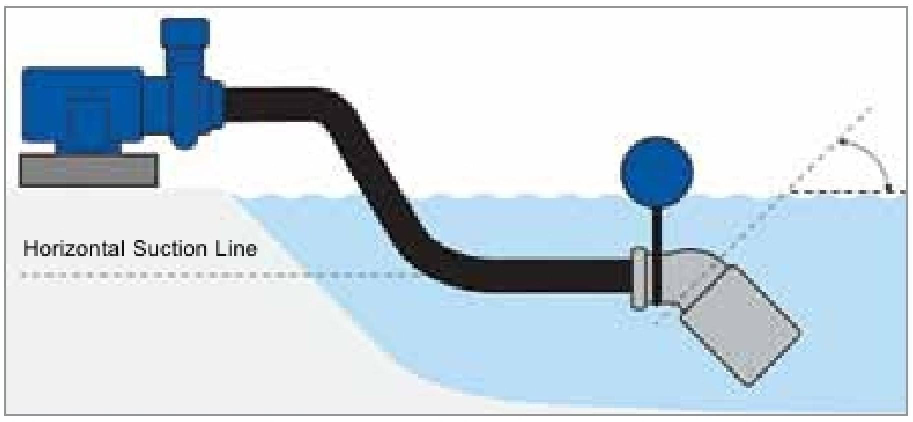

Fig. 7: Floating foot valve (Credit: http://sure-flo.com/foot-valves)

A floating valve can be installed at the inlet of the additional reserve/backup tank. It will stop incoming hot water when the reserve tank is full. The floating valve automatically drops down when the water level goes down.

The block diagram of the solar water heater system with the reserve tank and the floating valve is shown in Fig. 6. When hot water flows out from outlet of the tank, the cold water starts entering into the reserve tank. But during absence of sunlight or at night, the circuit prevents cold water from flowing into the solar water heater tank by blocking the main water inlet.

At the same time the available hot water will not be able to flow into the reserve tank. So, a floating foot valve is required to keep the outlet pipe sink in hot water.

Illustration of a typical floating foot valve is shown in Fig. 7. The floating ball will keep the outlet pipe submerged to ensure uninterrupted water flow.

After wiring the whole circuit, connect the power supplies (Fig. 3) to the main circuit (Fig. 2) via CON1 and CON2 for 5V and 12V, respectively. Turn on the power supplies.

The LDR1 will detect the intensity of light continuously. At day time, the signal at digital input pin 13 of the Arduino is low. The low signal at pin 13 will keep the relay (RL1) de-energised. So, the solenoid valve (SV) connected to NC contact of the relay will get energised, allowing cold water to flow into the solar water heater tank. The hot water is further stored in the reserve tank.

At night, the circuit will give a high signal to pin 13 of Arduino, which will energise the relay (RL1). This will disconnect the solenoid valve (SV) from the 12V power supply and stop the flow of cold water into the solar water heater tank. EFY

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